

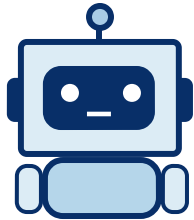
1

Prompt Routing

Identify mixed-outcome prompts



Solve for x:
 $2(x - 3) + 4 = 14$



Student model

Generate multiple rollouts

- ✓ ... → x = 8
- ✗ ... → x = 6
- ✗ ... → x = 12
- ⋮

All-success

- ✓ ... → x = 8
- ✓ ... → x = 8
- ✓ ... → x = 8
- ⋮

▶▶ Skip

No need to route
to teacher

Mixed-outcome
Both correct and incorrect

- ✓ ... → x = 8
- ✗ ... → x = 6
- ✗ ... → x = 12
- ⋮

▶ Route

Send to rollout
routing (Stage 2)

All-failure

- ✗ ... → x = 6
- ✗ ... → x = 12
- ✗ ... → x = 7
- ⋮

▶▶ Skip

No useful signal
for training

2

Rollout Routing

Select one failed rollout to query

Same-prompt success reference

r_1

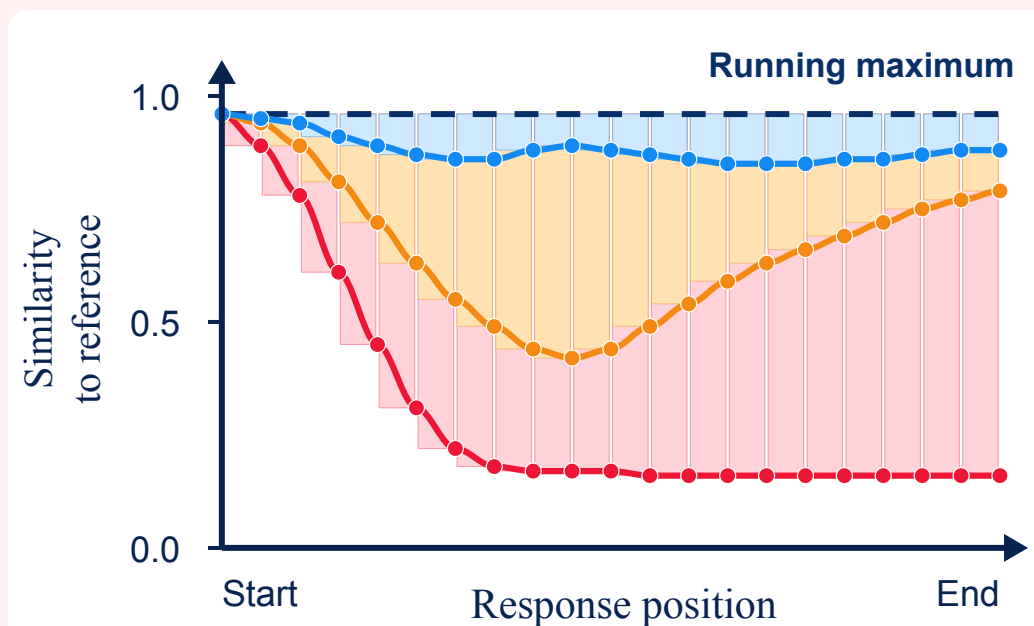


$$2x - 2 = 14 \rightarrow 2x = 16 \rightarrow x = 8$$



Reference only,
not sent to Teacher

Failed rollout candidates



--- Running maximum
Discrete departure area

Failed candidate (r_2)
✗ $2x - 2 = 14 \rightarrow \dots$
 $2x = 16 \rightarrow x = 6$
Small departure

Failed candidate (r_4)
✗ $2x - 10 = 14 \rightarrow \dots$
 $2x = 16 \rightarrow x = 7$
Medium departure

Failed candidate (r_3)
✗ $2x - 10 = 14 \rightarrow \dots$
 $x = 12$
Large departure



Selected failed rollout

$$2x - 10 = 14 \rightarrow \dots \rightarrow x = 12$$

r_3

Training Handoff for the Selected Rollout



Compute cost-aware score
Rank failed rollouts by
departure and token efficiency



Select one failed rollout
Use the highest-scoring
candidate



Query frozen Teacher
Get the target response
(stop-gradient)



Standard OPD update
Update the student model
with the selected pair